AgentWorkShop SuperAgent Architecture — Project as a governed superagent over channel sub-agents

Owner
human authority
sole sign-off

GATE ① plan approval

GATE ② result acceptance

GATE ③ project completion

re-checked at every decision

SUPERAGENT = Project

one governed agent — LLM proposes & verifies · deterministic side decides

LLM Hemisphere — Chief Leader

evaluate tasks · propose plans
suggest routes (gate default off)
L1 technical acceptance
propose replans

never: dispatch · self-approve · complete

Deterministic Hemisphere — Executor

Ready(W) predicate (TechOK · capacity)
atomic dispatch (CAS · no orphans)
retry / rework / deadlock detection
handoff assembly (contract + evidence)

every decision audited & reproducible

1
1
2

Channel A — Frontend

LLM lead + workers · tools
exec modes · handoff→B

Channel B — Backend

same governed envelope
relay set by executor only

Channels C…N — envelope

MEMORY & REFLEXES — dual ledger (decisions · events) · audit + DCW journal · effects outbox (retry → dead) · watchdog + HealthBoard

Quality sensor — three-level acceptance

per deliverable · L1 releases downstream · L2 releases completion

L0 Content Check
real artifact · sha256
→ content_ready

L1 Technical (Leader)
vs locked criteria
→ technical_accepted

L2 Business (Owner)
deliverable approved
→ releases completion

GATE ③ + L2 → project completed
never auto-closes → artifacts · audit

Guarantees — audited & reproducible
effectively-once + idempotence

Industry touch
bound channels add:
DAQ read → TSDB
DCW write via throat (HITL)
MES bridge · twin-first

details → Execution Flow figure

green = LLM hemisphere · red = no-LLM zone · orange dashed = human gates · dark = memory